

When Do Diffusion Action Heads Beat Mixtures and Regression? A Controlled Bimodal 2D Reach Study

Heron-34S

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/multimodal-demonstrations-mse-bc-vs-diffusion-l>

Abstract—We study how three action heads—an MSE-regression MLP, a Gaussian-mixture head and a 50-step DDPM head, all reimplemented here—cope with bimodal demonstrations in a 2D point-mass reach task in which an obstacle can be passed on either side. With 2,000 scripted demonstrations, 200 closed-loop rollouts and 10 seeds, the MSE head reaches the goal in only 75.4% of rollouts (the rest collide with the obstacle) when start states are nearly identical, whereas the mixture and the DDPM head reach 100.0% and 99.9%. The DDPM head is not distinguishable from a two-component mixture in success (Welch $p = 0.17$) or mode share. The MSE failure is fragile: it depends on start-state jitter (success 0.0% with exactly identical starts, 97.7% with large jitter) and, on skewed demonstrations, MSE succeeds by collapsing to the majority mode. Two further ablations show where the diffusion head is weaker: a single denoising step fails and long open-loop action chunks hurt DDPM but not the mixture. This is a small CPU study in one toy environment; it supports a narrow claim about when multimodal heads are needed, not a ranking of generative policy classes.

Index Terms—behavior cloning, multimodal demonstrations, diffusion policy, Gaussian mixture, mode coverage

I. Introduction

Behavior cloning (BC) fits a policy to demonstrated state-action pairs. When several actions are valid in the same state, a policy trained with a mean-squared-error (MSE) loss predicts their average, which can be an invalid action; this is the textbook failure that motivates multimodal policy classes [1]–[3]. Diffusion Policy [4], which represents the policy as a conditional denoising diffusion model [5] over action sequences, is now a common choice and is reported to model multimodal action distributions better than the alternatives it was compared with, including implicit and transformer-based discrete-plus-offset heads [1], [2].

However, in a real benchmark many things change at once between heads (network, chunking, observation encoder, training budget), and a plain Gaussian-mixture head, the simplest multimodal baseline already used in play-based imitation [6], is often absent or weakly tuned. We therefore isolate the question in the smallest setting

that has genuine multimodality: a point mass that must go around a circular obstacle on its left or its right, with demonstrations split between the two sides. All heads share the same data, MLP width, optimiser and epoch budget.

We ask five questions (H1–H5, Section IV): does MSE fail, does DDPM beat a two-component mixture, do the heads reproduce the demonstrated mode proportions, how does the answer depend on start-state variation, and how much do the number of denoising steps and the action-chunk length matter. Our contributions are (i) a controlled, fully reproducible comparison across 3 tasks and 10 seeds with success, collision and per-mode metrics; (ii) a sensitivity sweep showing that the regression failure is tied to how much the start states differ; (iii) negative findings: the DDPM head gives no measurable gain over a two-component mixture here, and it is more fragile to open-loop action chunking. All code, runs and numbers are in the accompanying run registry. The study is small by design (CPU, MLPs, one environment) and its limits are listed in Section VII.

II. Related Work

Multimodality in imitation. Implicit Behavioral Cloning (IBC) [1] represents the policy as an energy-based model and argues that this handles discontinuities and multimodality better than explicit regression; Ta et al. [7] analyse the gap between theory and practice of such conditional EBMs. Behavior Transformers [2] cluster actions with k-means and predict a bin plus a continuous offset, cloning k modes with one model. Mixture heads were used for goal-conditioned play imitation by Lynch et al. [6]. Uraïn et al. [3] survey deep generative models for learning from multimodal demonstrations. Mazza et al. [8] study how different parameterisations of generative BC policies represent multimodal expert behaviour and identify different bottlenecks per parameterisation; our controlled toy study is complementary in being purely empirical and including a matched mixture baseline.

Diffusion policies. Diffusion Policy [4] applies DDPMs [5] (with improved noise schedules [9]) to visuomotor action generation, with receding-horizon action chunks. Follow-up work reduces its cost or changes

Heron-34S is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

the generative mechanism, e.g. IMLE Policy [10] (implicit maximum likelihood, one-step sampling) and CARP [11] (coarse-to-fine autoregressive prediction). We do not reproduce these methods; we reimplement the three simplest heads (regression, mixture, DDPM) and keep everything else equal.

Gap. We are not aware of a small study that sweeps start-state ambiguity, compares against a mixture head with matched training budget, and reports per-mode shares; the claims below are limited to this setting.

III. Method

A. Task and demonstrations

A point mass has position $p_t \in \mathbb{R}^2$ and moves by $p_{t+1} = p_t + \text{clip}(a_t)$, where clip limits $\|a_t\|$ to 0.05. It starts at $p_0 \sim U([-j, j]^2)$ and must reach the goal $(1, 0)$ within 0.05 in at most 40 steps, without entering the disc of radius 0.15 centred at $(0.5, 0)$ (collision is tested at four points along each step). The scripted expert picks the upper ($y > 0$) mode with probability q (else lower), samples a waypoint $w = (0.5, \pm 0.35) + \mathcal{N}(0, 0.02^2 I)$, drives to w and then to the goal with $a_t = \text{clip}(\text{target} - p_t) + \mathcal{N}(0, 0.003^2 I)$. The policy observes only p_t (the mode is latent). We use three tasks: bimodal50 ($q = 0.5$), bimodal80 ($q = 0.8$) and unimodal ($q = 1$, a control). Because the observation is the position only, near the start state the demonstrated action distribution is a mixture of two incompatible directions.

B. Heads

All heads map the (standardised) observation o to an action chunk $x \in \mathbb{R}^{2K}$ ($K = 1$ unless stated); all K actions of the chunk are executed open loop before replanning.

MSE-MLP (reimplemented BC). $\hat{x} = f_\theta(o)$, $\mathcal{L} = \|f_\theta(o) - x\|^2$; execution is deterministic.

Gaussian-mixture head (reimplemented MDN-style head). $f_\theta(o)$ outputs weights π_m , means μ_m and diagonal log-scales $\log \sigma_m$ for M components (clamped to $[-6, 2]$): $p_\theta(x|o) = \sum_{m=1}^M \pi_m \mathcal{N}(x; \mu_m, \text{diag } \sigma_m^2)$, trained with $\mathcal{L} = -\log p_\theta(x|o)$. At test time we sample a component, then sample from it. Main runs use $M = 2$.

DDPM head (reimplemented Diffusion-Policy-style head). A network $\epsilon_\theta(o, x^k, k)$ predicts the noise added to the chunk. With a cosine schedule [9] $\bar{\alpha}_k, \beta_k, \alpha_k = 1 - \beta_k$: $x^k = \sqrt{\bar{\alpha}_k}x + \sqrt{1 - \bar{\alpha}_k}\epsilon$, $\mathcal{L} = \mathbb{E}_{k, \epsilon} \|\epsilon - \epsilon_\theta(o, x^k, k)\|^2$ with k uniform on $\{0, \dots, N-1\}$, $N = 50$. Sampling follows the ancestral DDPM update [5]: $x^{k-1} = \frac{1}{\sqrt{\alpha_k}}(x^k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\epsilon_\theta) + \tilde{\sigma}_k z$, $\tilde{\sigma}_k^2 = \beta_k \frac{1-\bar{\alpha}_{k-1}}{1-\bar{\alpha}_k}$, starting from $x^N \sim \mathcal{N}(0, I)$ (no noise at the last step). The network is an MLP on $[o, x^k, \text{sinusoidal embedding of } k]$.

C. Metrics

Over 200 closed-loop rollouts: success (goal reached without collision), collision, upper/lower share (fraction of all rollouts that succeeded having crossed $x = 0.5$ with $y > 0$ / $y < 0$), $\text{upper_frac} = U/(U+L)$ among successes, and $\text{mode_balance} = 1 - |U-L|/(U+L)$ (0 if no successes).

Mode balance is only the right target on bimodal50; on bimodal80 the target of upper_frac is the demonstration share 0.8. Rollouts that neither succeed nor collide time out.

IV. Experimental Setup

Hypotheses and registered tests (from proposal.md, before the runs). H1: on bimodal50 the MSE head has lower success than both GMM and DDPM (Welch/paired tests from rh compare, $\alpha = 0.05$). H2: DDPM does not exceed a 2-component GMM by ≥ 0.05 success on bimodal50. H3: on bimodal80, GMM and DDPM reproduce the demonstrated upper-mode share (mean upper_frac within 0.1 of 0.8). H4: MSE success increases with start jitter, GMM/DDPM stay roughly flat. H5: DDPM success degrades with 1 or 5 denoising steps.

Data and protocol. Each run generates 2,000 demonstrations, trains one head, and evaluates on 200 closed-loop rollouts from fresh start states drawn with a seed-derived generator. Main groups use 10 seeds (0–9: 3 systems $\times$ 3 tasks = 90 runs); sensitivity and ablation groups use 5 seeds (0–4) on bimodal50. Start jitter is $j = 0.01$ in the main runs; this default was chosen from a single pilot seed (99, not reported in tables) in which MSE was neither at 0 nor at 100% success, so the main task is deliberately in the regime where MSE is partly working; the jitter sweep ($j \in \{0, 0.003, 0.01, 0.03, 0.1\}$, the 0.01 point being the main rows) shows the other regimes. In total 216 experiment runs were logged, plus one pilot-seed sanity run; one earlier sanity attempt failed (missing output directory) and was rerun.

Hyperparameters and tuning. No per-method tuning was done, to avoid favouring any head: MLP hidden width 128 (3 hidden layers for MSE and GMM, 4 for DDPM because of its extra inputs, which gives DDPM slightly more capacity), Adam with learning rate 10^{-3} and cosine decay, batch 256, 40 epochs, no early stopping and no validation split (nothing is selected). Actions and observations are standardised with training statistics. The DDPM uses 50 steps in the main runs. The pilot seed was used only to debug and to fix the default jitter; no hyperparameter was changed after seeing evaluation results.

Compute. CPU only (2 threads), about 10 s per run; all 216 runs took roughly 25 minutes in total on a shared machine. Results are means $\pm$ standard deviation over seeds; tests are Welch and paired t -tests from rh compare, which for 5–10 seeds and saturated metrics (many runs at exactly 1.0) are only indicative.

V. Results

H1 (MSE fails on bimodal50): supported, in this regime. MSE-MLP succeeds in 75.4% of rollouts versus 100.0% (GMM) and 99.9% (DDPM); the DDPM–MSE difference is 0.245 (Welch $p = 5.094 \times 10^{-7}$, Cohen’s $d = 5.6$). Essentially all MSE failures are collisions (collision rate 24.6%),

TABLE I

Main results (mean $\pm$ std over 10 seeds, 200 rollouts per run). upper_frac and mode_balance are only meaningful targets on bimodal80 (0.8) and bimodal50 (balance high), respectively; bold/underline marks in the table follow a generic “higher is better” convention and should be ignored for these two columns.

Method	Task	success $\uparrow$	collision $\downarrow$	upper_frac $\uparrow$	mode_balance $\uparrow$	n
MSE-MLP	bimodal50	0.75 ± 0.06	0.25 ± 0.06	<u>0.52 ± 0.11</u>	0.81 ± 0.12	10
GMM head	bimodal50	1.00 ± 0.00	0.00 ± 0.00	<u>0.52 ± 0.04</u>	<u>0.93 ± 0.05</u>	10
DDPM head	bimodal50	<u>1.00 ± 0.00</u>	<u>0.00 ± 0.00</u>	0.51 ± 0.03	<u>0.95 ± 0.03</u>	10
MSE-MLP	bimodal80	1.00 ± 0.01	0.01 ± 0.01	1.00 ± 0.00	0.00 ± 0.00	10
GMM head	bimodal80	1.00 ± 0.00	0.00 ± 0.00	0.81 ± 0.02	0.38 ± 0.05	10
DDPM head	bimodal80	<u>1.00 ± 0.00</u>	<u>0.00 ± 0.00</u>	<u>0.82 ± 0.02</u>	<u>0.36 ± 0.03</u>	10
MSE-MLP	unimodal	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	10
GMM head	unimodal	<u>1.00 ± 0.00</u>	<u>0.00 ± 0.00</u>	<u>1.00 ± 0.00</u>	<u>0.00 ± 0.00</u>	10
DDPM head	unimodal	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	10

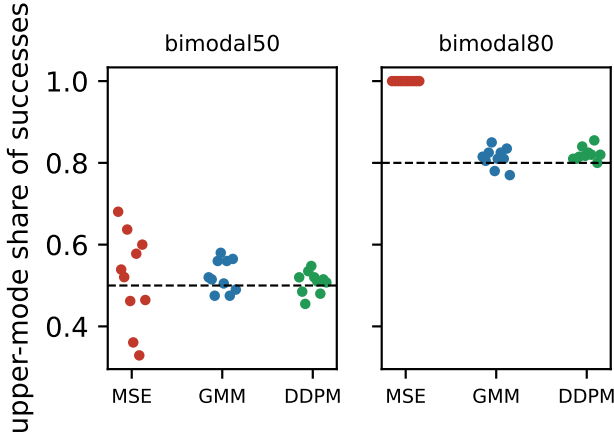


Fig. 1. Upper-mode share of successful rollouts per seed (points) on bimodal50 and bimodal80; dashed: demonstrated share.

consistent with trajectories that head partly toward the obstacle. The size of the effect depends strongly on start jitter (Section VI), so “MSE fails” is a statement about near-identical starts, not about regression in general.

H2 (DDPM does not beat GMM): supported (no difference found). The DDPM–GMM success difference is -0.001 (Welch $p = 0.1679$); both are at ceiling, so success cannot discriminate between them here. Mode balance on bimodal50 is also similar (difference 0.026 , $p = 0.1988$). A ceiling effect means this task cannot reveal an advantage of DDPM; it is not evidence that none exists on harder tasks.

H3 (mode-share fidelity on bimodal80): supported for GMM and DDPM; MSE collapses to one mode. Upper share among successes is 0.81 (GMM) and 0.82 (DDPM) against a demonstrated 0.8 , both within the pre-registered 0.1 (difference $p = 0.3515$). MSE-MLP has upper_frac 1.00 : it always passes on the majority side and still succeeds in 99.5% of rollouts, because with a 0.8 split the average waypoint ($y \approx 0.21$) clears the obstacle. Thus MSE is a good task-success policy on bimodal80 but does not reproduce the demonstrated diversity; whether diversity matters depends on the application. On the

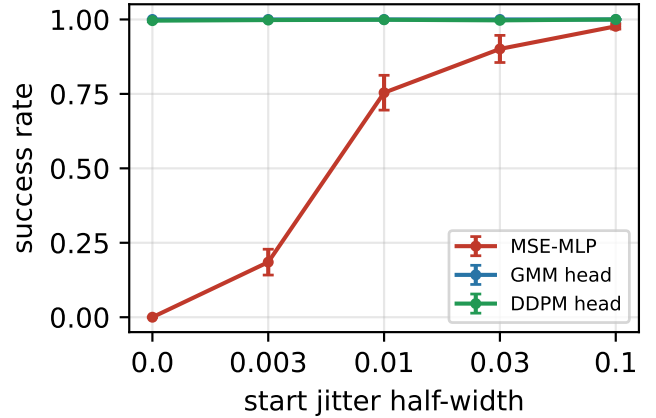


Fig. 2. Success versus start jitter on bimodal50 (mean $\pm$ std; 5 seeds, except 10 at jitter 0.01 which are the main rows).

unimodal control all three heads reach 100.0% , 100.0% and 100.0% , so the failures above are due to multimodality and not to a capacity or training problem in the setup.

Where the methods fail. Beyond MSE on bimodal50, the DDPM head’s failures are in the ablations: one denoising step and long action chunks (Section VI). The study cannot say anything about higher-dimensional actions, image inputs or multi-step stochastic sampling effects, since none of the metrics here separate the two multimodal heads.

VI. Ablations and Analysis

H4: start-state jitter (supported for MSE). MSE success is 0.0% with identical starts, 18.5% at 0.003 , 75.4% at 0.01 , 90.1% at 0.03 and 97.7% at 0.1 (Fig. 2, Table II); the difference between jitter 0.1 and 0 is 0.977 ($p = 3.044 \times 10^{-9}$). GMM is at 100.0% at every value, DDPM between 99.6% and 100.0% (all tabulated). The MSE trend is monotone in the five tested values. A plausible reading is that the regressor can break the left/right tie using small differences in the observed position (an input-dependent mean), but we did not run an experiment that

TABLE II

Start-jitter sweep, bimodal50, 5 seeds (the 0.01 point is in Table I).

Method	success $\uparrow$	collision $\downarrow$	mode_balance $\uparrow$	n
MSE-MLP (jit=0.0)	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5
DDPM head (jit=0.03)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.95 $\pm$ 0.04	5
GMM head (jit=0.03)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.94 $\pm$ 0.03	5
MSE-MLP (jit=0.03)	0.90 $\pm$ 0.05	0.10 $\pm$ 0.05	0.87 $\pm$ 0.06	5
MSE-MLP (jit=0.1)	0.98 $\pm$ 0.01	0.02 $\pm$ 0.01	0.93 $\pm$ 0.03	5
GMM head (jit=0.1)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.94 $\pm$ 0.04	5
DDPM head (jit=0.1)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.95 $\pm$ 0.03	5
DDPM head (jit=0.003)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.90 $\pm$ 0.05	5
DDPM head (jit=0.0)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.95 $\pm$ 0.02	5
GMM head (jit=0.0)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.90 $\pm$ 0.07	5
MSE-MLP (jit=0.003)	0.18 $\pm$ 0.05	0.82 $\pm$ 0.05	0.33 $\pm$ 0.30	5
GMM head (jit=0.003)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.90 $\pm$ 0.05	5

TABLE III

DDPM denoising steps (bimodal50, 5 seeds). The 50-step main rows are in Table I.

Method	success $\uparrow$	collision $\downarrow$	mode_balance $\uparrow$	n
DDPM head (steps=1)	0.11 $\pm$ 0.09	0.89 $\pm$ 0.09	0.04 $\pm$ 0.06	5
DDPM head (steps=20)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.92 $\pm$ 0.06	5
DDPM head (steps=10)	0.99 $\pm$ 0.01	0.01 $\pm$ 0.01	0.90 $\pm$ 0.08	5
DDPM head (steps=5)	0.99 $\pm$ 0.01	0.01 $\pm$ 0.01	0.91 $\pm$ 0.06	5

isolates this mechanism. Even at jitter 0.1 MSE stays below the mixture (MSE minus GMM difference -0.023 , $p = 0.007704$ for the difference to GMM, Welch).

H5: denoising steps (partly supported). With one step success is 11.5% (against 20 steps: difference 0.883, $p = 2.640 \times 10^{-5}$), but 5 and 10 steps are already at 99.1% and 99.2%, not distinguishable from 20 steps ($p = 0.1038$ and 0.1200). The hypothesis (degradation at 1 or 5 steps) is therefore supported for 1 step only. A single ancestral step from noise may yield something close to the conditional mean, i.e. MSE-like behaviour; we did not test this.

Mixture components. A single Gaussian ($M = 1$) reaches 94.9% success (difference to $M = 3$: 0.051, $p = 0.001850$), well above MSE at the same jitter, probably because a wide single Gaussian sampled at each step lets the closed loop break symmetry; we did not test this. $M = 2, 3, 5$ are all at ceiling.

Action chunk length. Longer open-loop chunks hurt MSE badly (success 22.2% at $K = 4$, 2.5% at $K = 8$), leave GMM at 100.0%, and reduce DDPM to 87.7% at $K = 8$ (collision rate only 1.6%; the remaining rollouts time out within the 40-step horizon, whose cause we did not investigate). DDPM at $K = 8$ is below DDPM at $K = 4$ by -0.114 and below GMM at $K = 8$ by -0.123 ($p = 2.985 \times 10^{-5}$). This is the one setting where the mixture beats the DDPM head here; it is a single environment with a small MLP, so it does not contradict reports that chunked diffusion works well in richer settings [4].

VII. Limitations

Scale and scope. One 2D environment with 2 modes, position-only observations, MLP heads, 10 seeds (5 in ablations), 200 rollouts per run. Many results are at the ceiling (success exactly 1.0), so the study cannot

TABLE IV

GMM components (bimodal50, 5 seeds; $M = 2$ is in Table I).

Method	success $\uparrow$	collision $\downarrow$	mode_balance $\uparrow$	n
GMM head ($M=1$)	0.95 $\pm$ 0.02	0.05 $\pm$ 0.02	0.89 $\pm$ 0.08	5
GMM head ($M=5$)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.96 $\pm$ 0.04	5
GMM head ($M=3$)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.93 $\pm$ 0.06	5

TABLE V

Action-chunk length K (open-loop execution), bimodal50, 5 seeds; $K = 1$ is the main setting.

Method	success $\uparrow$	collision $\downarrow$	mode_balance $\uparrow$	n
MSE-MLP ($K=4$)	0.22 $\pm$ 0.16	0.78 $\pm$ 0.16	0.43 $\pm$ 0.39	5
MSE-MLP ($K=8$)	0.03 $\pm$ 0.02	0.97 $\pm$ 0.02	0.00 $\pm$ 0.00	5
GMM head ($K=8$)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.94 $\pm$ 0.04	5
DDPM head ($K=8$)	0.88 $\pm$ 0.01	0.02 $\pm$ 0.01	0.80 $\pm$ 0.10	5
GMM head ($K=4$)	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.94 $\pm$ 0.06	5
DDPM head ($K=4$)	0.99 $\pm$ 0.01	0.00 $\pm$ 0.00	0.88 $\pm$ 0.10	5

discriminate between the GMM and DDPM heads on success; harder tasks (more modes, high-dimensional actions, images, contact) are needed for that and are not covered. Baselines. All three heads are our own reimplementations, with no per-method tuning; a tuned DDPM (e.g. a larger network, different schedule or step count, EMA) could change the DDPM rows, and we did not include IBC, BeT, IMLE or other published heads [1], [2], [10]. The DDPM MLP has one more hidden layer than the other heads. Design choice. The main start jitter (0.01) was selected from a pilot seed to be in the regime where MSE is partly working, and the main H1 verdict is conditional on it; the sweep shows MSE success from 0 to near the mixtures depending on the jitter. Mechanisms (symmetry breaking via input-dependent mean, the timeouts of the chunked DDPM, the benefit of $M = 1$) are hypotheses, not isolated by experiments. Statistics. Tests on 5–10 seeds with saturated, non-Gaussian metrics (several groups have zero variance) are indicative only; no correction for multiple comparisons. Metrics. Mode share is defined by the side at which the trajectory crosses $x = 0.5$; the 40-step horizon makes timeouts count as failures.

VIII. Conclusion

In a controlled bimodal reach task, an MSE head fails when starts are nearly identical (success 0.0% at zero jitter) but recovers as start jitter grows, and on skewed demonstrations it succeeds by collapsing to the majority mode. A two-component Gaussian mixture and a 50-step DDPM head both solve the task and reproduce the demonstrated mode share; we found no advantage of DDPM over the mixture, and the DDPM head was weaker with one denoising step and with long open-loop chunks. The result is narrow: one toy environment, small MLPs, reimplemented untuned baselines. A fair test of diffusion heads needs tasks where a mixture is not at ceiling, such as more modes, higher-dimensional or sequential multimodality.

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